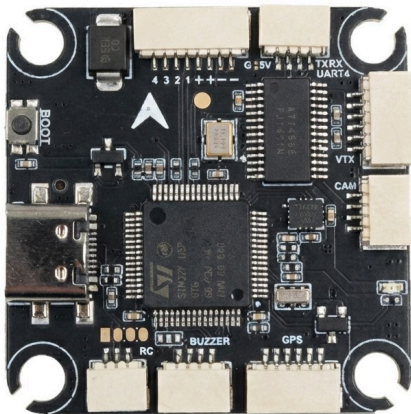
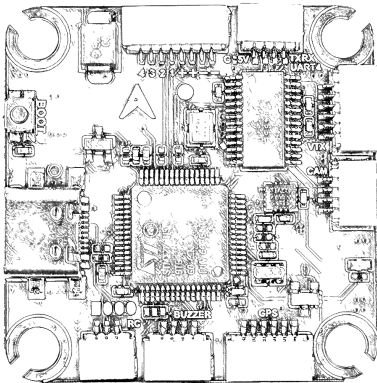


AF-F4 nano

User Guide



AF-F4 nano Overview



Compact STM32F405 flight controller with an ICM-42688-P IMU. The novaX hardware definition configures an SPL06 barometer, 8 motor/servo outputs, 1 LED output and 6 UARTs plus USB.

AF-F4 nano Specifications

• Basic Specifications

MCU	STM32F405
IMU	ICM-42688-P
Barometer	SPL06 (current novaX hardware definition)
Operating Voltage	9 – 25 V DC
Output Voltage	3.3V/1A · 5V/3A · 10V/3A
USB Input	USB Type-C (Firmware & Power)
PWM Output	8 motor/servo channels + 1 LED channel
Serial Ports	6 hardware UARTs + USB configured; verify connector pinout

AF-F4 nano Specifications

• Basic Specifications

Serial Mapping	SERIAL1 = USART1 · SERIAL2 = USART2 · SERIAL3 = USART3 · SERIAL4 = UART4 · SERIAL5 = UART5 · SERIAL6 = USART6 (USB = SERIAL0)
Servo Output	2 Port
RC Input	PWM / PPM / S.Bus
Size	39.4 × 39.4 mm
Mount Hole	30.5 × 30.5 mm / M4
Weight	9.3 g
Operating Temp	-20 ~ +70 °C
Supported F/W	novaX ArduPilot (released); Betaflight board configuration available

AF-F4 nano Pinout

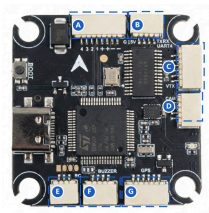
• Pinout (1/4)

AF-F4 nano Pinout

Both sides seen straight on. Where the board silk names each pin (ESC, ELRS and every bottom pad) the table follows that silk in the order you see it in the picture next to it; the other top connectors keep the order of the maker's drawing (marked on each card). There are no pin numbers.

A = connector or pad group, its table below red = power black = GND blue = signal grey = not connected

Top side



A ESC

8-Pin · motor signals S1–S4 + battery · board silk 4 3 2 1 + + - - (= VBAT)

S4	S3	S2	S1	+	+	-	-
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B ELRS

4-Pin · UART4 receiver · board silk G 5V TX RX

GND	5V	TX4	RX4
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AF-F4 nano Pinout

• Pinout (2/4)



C VTX

6-Pin · UART3 TX · video out · order from the maker's drawing (no silk per pin)

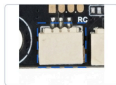
NC
GND
NC
NC
VTX
TX3



D CAM

3-Pin · analog video in · order from the maker's drawing (no silk per pin)

5V
GND
CAM



E RC

3-Pin · PPM receiver input · order from the maker's drawing (no silk per pin)

GND	5V	PPM
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F BUZZER

4-Pin · UART3 RX · buzzer output · order from the maker's drawing (no silk per pin)

5V	RX3	BUZZ	GND
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G GPS

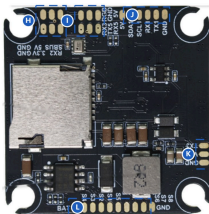
6-Pin · UART1 + I2C1 · GNSS / compass · order from the maker's drawing (no silk per pin)

SDA1	SCL1	RX1	TX1	5V	GND
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AF-F4 nano Pinout

• Pinout (3/4)

Bottom side (pads)



H SBUS pads

UART2 RX · RC input — 2 × 3 pads · the silk below is printed upside down

GND	5V	SBUS
GND	3.3V	RX2

I RGB pads

UART3 / UART5 · LED strip or serial — 2 × 3 pads

RGB	GND	5V
RX3	TX5	RX5

J I2C / UART pads

I2C1 / UART1 · external bus

5V	SDA1	SCL1	RX1	TX1	GND
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AF-F4 nano Pinout

• Pinout (4/4)



K VTX pads

UART3 TX - VTX control

TX3
VTX
GND



L MOTOR / POWER pads

Motor signals S1-S8 + battery

BAT	S4	S3	S2	S1	S5	S6	S7	S8	GND
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Also on the board: MCU STM32F405RG (168 MHz · 1 MB flash · 192 KB RAM) · IMU ICM-42688-P · OSD AT7456E · BOOT button and USB Type-C on the top · TF card slot and the BEC on the bottom.

AF-F4 nano Setup

1. **Mounting** — Align the board's forward direction with the vehicle nose on a vibration-damped mount; otherwise set AHRS_ORIENTATION. Mounting: 30.5 × 30.5 mm / M4.
2. **Power** — Supply power within the operating range and check polarity first. Operating voltage: 9 – 25 V DC.
3. **Peripherals** — Connect the receiver, GPS / compass and ESCs according to the pinout.
4. **Firmware** — Load this product's novaX ArduPilot image from the catalog Web Updater; never use another board's firmware.
5. **Setup** — In Mission Planner or similar, choose the frame, calibrate accelerometer, compass and radio, and set flight modes and failsafes.

AF-F4 nano Firmware

• Available Firmware

Type	Version	Date
ArduPilot (.apj package)	1.2.3	2026-07-11
Bootloader + App (merged HEX · DFU / SWD)	1.2.3	2026-07-11

Download files from the product page (QR).

AF-F4 nano Safety Notes

- Always remove propellers while wiring, configuring or updating firmware.
- Pull connectors by the housing, never by the wires.
- Beware of static discharge; do not operate in wet or dusty conditions.



<Detailed Manual>

